

Trip Hawks / AirAgent

Full-Stack System Analysis, Technical Architecture, and Product Requirements Document (PRD)

DOCUMENT TYPE	VERSION & STACK	STATUS
PRD & Architecture	Next.js 16 + Express 5 + Mongo	Production Specification

1. Executive Summary

Trip Hawks (AirAgent) is an enterprise-grade, modern omnichannel travel technology platform engineered to facilitate domestic and international flight search, hotel bookings, interstate bus tickets, car rentals, and curated vacation packages.

The platform merges high-speed self-service booking capabilities with a conversational **Google Gemini 2.5 AI Travel Concierge** and an offline assisted-booking desk, backed by transactional automated email dispatch workflows.

Core Strategic Differentiator

Combines live GDS inventory (Flyshop API) with an assisted booking lead engine, allowing customers to either book directly online or connect directly with specialized travel consultants via dedicated hotlines and assisted lead triggers.

2. System Architecture & Tech Stack

The codebase is organized as a decoupled monorepo featuring a TypeScript-driven full-stack ecosystem:

Domain	Framework / Technology	Architectural Role & Description
Frontend	Next.js 16.2.9, React 19.2.4, TailwindCSS v4	Server & Client components, App Router, reactive autocomplete, dark mode state persistence, Framer Motion animations.
Backend API	Express 5.2.1, TypeScript 5.8, Node.js (ESM)	Stateless REST API with esbuild bundling, async error boundaries, schema validation with Zod.
Database & Data	MongoDB & Mongoose 9.1.6	Document store with strict typing for User records, Flight Booking Requests, and multi-currency conversion tables.
AI Services	Google Gemini 2.5 Flash (@google/genai)	Conversational travel assistant trained on custom system instructions with bilingual (English & Hinglish) support.
Authentication	Clerk SDK, Google OAuth, JWT, BcryptJS	Tri-mode auth supporting Clerk SSO, Google One-Tap, and self-hosted credential JWT auth with HTTP-only cookies.
Communications	Resend API (resend)	Automated HTML email generation for booking confirmations, customer itineraries, and instant admin escalation alerts.
Security	Helmet, Rate-Limit, Mongo-Sanitize	NoSQL injection defense, CSP headers, IP rate limiting, and structured logging via Pino.

3. Core Functional Modules

✈️ Flight Booking & Lead Engine

Features origin/destination IATA code resolution, passenger class selection (Economy, Premium, Business, First), real-time pricing, in-memory route caching, and instant booking modal with transactional email triggers.

FLYSHOP GDS INSTANT LEAD FLOW

🏨 Hotel & Accommodation Engine

Comprehensive hotel search with location autocomplete, check-in/out calendars, guest counters, star rating filters, amenity chips (WiFi, Pool, Breakfast), and detailed room breakdown modals.

CURATED INVENTORY DYNAMIC FILTERS

🗣️ Gemini 2.5 AI Travel Concierge

Interactive floating assistant equipped with step-by-step guidance, rate-limit retry logic (429/500/503), intent recognition, and bilingual English/Hinglish travel suggestions.

GEMINI 2.5 BILINGUAL SUPPORT

🚌 Buses, Car Rentals & Cruises

Interstate bus search with seat selection visualization, vehicle rental catalog, and international cruise holiday packages with promotional badge indicators.

MULTI-VERTICAL SEASONAL OFFERS

4. Data Model Specifications

Primary MongoDB collections managed through Mongoose schemas:

```
// Flight Booking Request Schema (IFlightBookingRequest) { requestId: { type: String, unique: true, index: true }, customer: { name: { type: String, required: true }, email: { type: String, required: true }, mobile: { type: String, required: true } }, flight: { airline: String, flightNumber: String, origin: String, destination: String, departureTime: String, travelDate: String, travelClass: String, price: Number, currency: String, stops: Number }, passengers: [{ paxType: "Adult" | "Child" | "Infant", title: String, firstName: String, lastName: String, gender: "Male" | "Female" | "Other", passportNumber: String }], status: "PENDING" | "CONTACTED" | "CONFIRMED" | "CANCELLED", createdAt: Date, updatedAt: Date }
```

5. Security, Reliability & Performance Audit

- **Defense-in-Depth:** Express 5 pipeline includes helmet for secure headers, express-mongo-sanitize for NoSQL injection prevention, and strict origin CORS whitelisting.
- **High Reliability AI Layer:** aiService.ts includes exponential backoff retry algorithms for Gemini API transient rate limits and server availability.
- **SEO & Crawler Optimization:** Implements Next.js dynamic sitemap.ts and robots.ts, pre-rendering meta tags for OpenGraph and Twitter cards.

6. Product Requirements Document (PRD) Matrix

Req ID	Module	Requirement Specification	Priority	Status
FR-01	Flights	IATA airport search, multi-city/roundtrip query, cabin filters.	P0	IMPLEMENTED
FR-02	Lead Capture	Booking modal with passenger details & instant dual email dispatch.	P0	IMPLEMENTED
FR-03	Hotels	Destination search, star rating filters, amenity selection & room modal.	P0	IMPLEMENTED
FR-04	AI Concierge	Gemini 2.5 chat with retry mechanisms & bilingual support.	P1	IMPLEMENTED
FR-05	Auth System	Clerk SSO + Google OAuth + Native JWT credential auth.	P1	IMPLEMENTED
FR-06	Payments	Online payment gateway integration (Stripe / Razorpay) for instant e-ticketing.	P1	ROADMAP
FR-07	User Portal	Self-service dashboard to track booking statuses, download e-tickets, and cancel.	P2	ROADMAP
FR-08	Admin Panel	Agent console to transition booking states (PENDING to CONFIRMED) & manage inventory.	P2	ROADMAP

7. Strategic Roadmap & Future Milestones

Phase 1: Payment Gateway Integration

Integrate Stripe & Razorpay Webhooks to enable zero-touch online checkout and automated PNR ticket generation upon successful capture.

Phase 2: Customer Account Portal

Develop customer profile pages with historical booking records, printable PDF boarding passes, and automated flight delay notifications.

Phase 3: Back-Office CRM & Admin UI

Create a secure dashboard for travel agents to assign lead ownership, trigger follow-up SMS/WhatsApp messages, and track sales revenue.

Phase 4: Distributed Redis Caching

Migrate in-memory route cache to a managed Redis cluster to optimize multi-node scale and reduce GDS query overhead.

Document Verification & Sign-off

This technical and product requirements document accurately reflects the existing architecture and future expansion specifications for the Trip Hawks / AirAgent project.